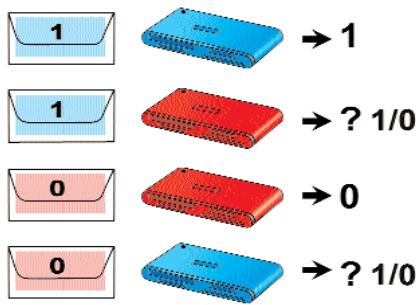


Ekert's Card Tricks



The Blue machine only understands blue cards—it makes a random guess with Red. Ditto the Red machine—it only understands red cards

the red machine, it will make a guess at what's on the card—say 0. But now, if this card is put back into the red machine, it will still read 0. But wasn't it supposed to give a random answer each time? What has happened is that the card has now taken on the properties of a red card with 0 written on it, and its ever being blue with 1 is now a distant memory. This is a manifestation of uncertainty: observing a card as blue automatically destroys the assumption that it was ever red, and vice versa.

Quantum Weirdness And Spooky Happenings

If you thought all this was bizarre, read on to find out what even the greatest of scientists call weird. It's called *Quantum Entanglement*, and had been billed by Einstein himself as "spooky."

The concept of quantum entanglement talks about photons that are so closely related that it's impossible to describe one without describing the other, no matter how far apart they are. And the really spooky part is that any attempt to measure the properties of one photon will immediately affect the other one—photons talking to each other in no time at all.

Coming back to the cards: suppose, now, that we hand one sealed envelope each to Alice and Bob. We find that if both of them use the same colour machine, their outputs are the same—either both 0 or both 1. But if they use different machines, the outputs may or may not be the same. But how? There is no way for both of them to know what colour card they are receiving. Let's try to guess what the card could be before either of them checks it in their respective machines. What if both cards were blue? That would surely get them the same answer if they both put the cards in their blue machines, but that doesn't mean that they would get the same if they put the card in their red machines. By similar reasoning, we can also discount that they were both red.

What if both were different colours? Even that wouldn't explain why both in a blue machine would give the same outcome—what happened to the logic of having random outcomes when the colours didn't match? The

only option left is that there was no colour or data on those cards before they were put in the machine.

What we didn't tell you is that the cards we gave Alice and Bob were really entangled—so if Alice's blue machine said that the card has 1 written on it, then *both* cards are blue with 1 written on it. Now, if Bob puts the card in his blue machine, it will (correctly) tell him that its value is 1, but if he puts it in his red machine, he will get either a 0 or 1 randomly—just as expected.

Why Won't You Talk To Me?

Finally, now, we are in a position to discuss how keys are exchanged in the quantum cryptography scenario.

We have our old favourites—Alice, Bob and Eve; joining them is a source of entangled photons—let's call him Sam. We can use the card analogy here as well.

Sam sends out pairs of entangled cards to both Alice and Bob, who decide, independently of each other, which machine to use to read it. Suppose Alice starts with her red machine and gets a reading of 0, and Bob uses his blue machine and gets the same reading. This goes on for a long while. After a set number of such trials, they start to compare notes. They discuss the measurements (but not the data they got from them) over a public channel—something like:

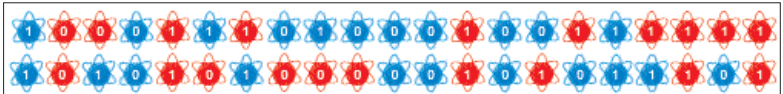
Alice: "I used my red machine."

Bob: "Hey so did I!"

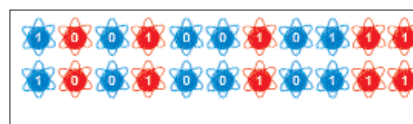
OR, Bob: "Nah, I used blue."

This continues for all the cards that were sent out by Sam. Finally, they keep only the data they got from those cards for which they made the same measurement, and discard the rest. Thanks to Quantum Weirdness, they know that they now have the same string of data. To make doubly sure, though, they share some of this data, again over the public channel, to see if

Getting The Key

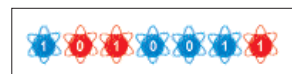
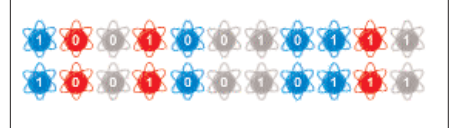


The measurements that Alice (top row) and Bob (bottom row) make. They get the same result when they make the same measurement.



Now that they've made their measurements, they only keep the results where they both used the same machine

They pick random results and compare them to see if there were errors



Finally, the Key!